



Process Planning and CAPP

Groover M.P.: Chapter 36



The student knows:

- What process planning is;
- What possible approaches to automation are.





Process planning is usually the principal function within the Manufacturing Engineering Department of a company.

It refers to planning the manufacturing processes for economical production of high quality products:

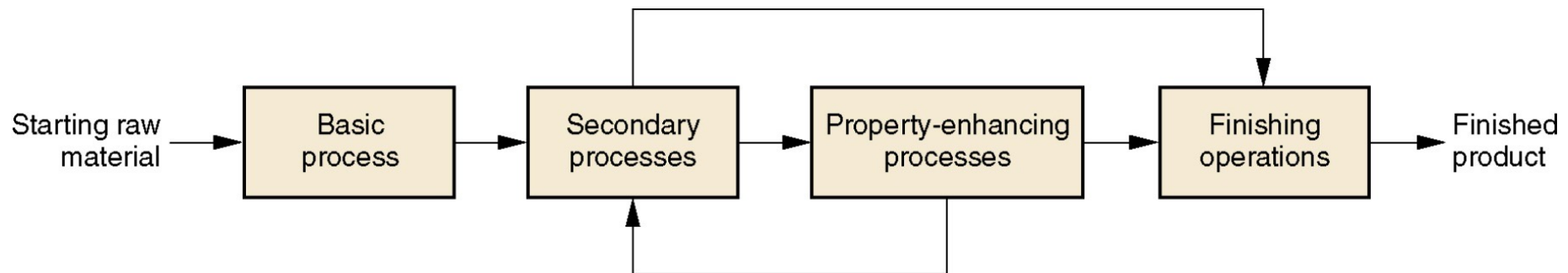
- Deciding appropriate **processes** and their **sequence**;
- Determining **tooling** requirements;
- Selecting **equipment**;
- Estimating **costs**.



TYPICAL PROCESSING SEQUENCE

A typical processing sequence to fabricate a discrete part consists of:

1. A basic process to establish initial geometry of part;
2. One or more secondary processes to transform starting geometry into final part shape;
3. Operations to enhance physical properties;
4. Finishing operations.





DETAILS IN PROCESS PLANNING

- **Processes and sequence** - process plan should briefly describe all processing steps used on the work unit in the order they are performed.
- **Select equipment** to be used - try to develop process plans that utilize existing plant equipment, otherwise, part must be purchased, or new equipment must be installed in the plant.
- **Specify** what tools, dies, molds, fixtures, and gages are needed.



DETAILS IN PROCESS PLANNING

- **Specify methods** - hand and body motions, workplace layout, small tools, hoists for lifting heavy parts:
 - Methods must be specified for manual operations and manual portions of machine cycles.
- **Estimate production costs** - often accomplished by working with cost estimators.
- Specify **cutting tools and cutting conditions** (speeds and feeds) for machining operations.



The document that specifies the details of the process plan:

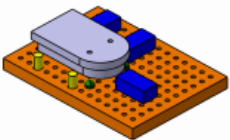
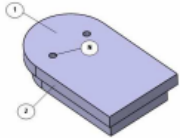
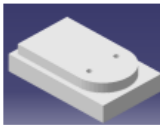
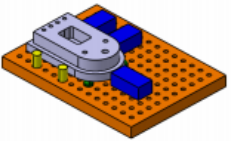
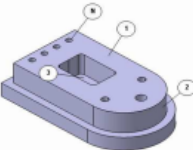
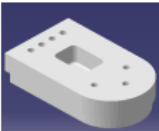
- The route sheet is to the process planner what the engineering drawing is to the product designer;
- Route sheet should include all manufacturing operations to be performed on the workpart, listed in the order in which they are to be performed.

Part No: 031393	Part Name: Housing, valve	Rev. 2	Page <u>1</u> of <u>2</u>			
Matl: 416 Stainless	Size: 2.0 dia × 5. long	Planner: MPG	Date: 3/13/XX			
No.	Operation	Dept.	Machine	Tooling, gages	Setup time	Cycle time
10	Face; rough & finish turn to 1.473 ± 0.003 dia. $\times 1.250 \pm 0.003$ length; face shoulder to 0.313 ± 0.002 ; finish turn to 1.875 ± 0.002 dia.; form 3 grooves at 0.125 width $\times 0.063$ deep.	L	325	G857	1.0 h	8.22 m
20	Reverse; face to 4.750 ± 0.005 length; finish turn to 1.875 ± 0.002 dia.; drill $1.000 + 0.006, -0.002$ dia. axial hole.	L	325		0.5 h	3.10 m
30	Drill & ream 3 radial holes at 0.375 ± 0.002 dia.	D	114	F511	0.3 h	2.50 m
40	Mill 0.500 ± 0.004 wide $\times 0.375 \pm 0.003$ deep slot.	M	240	F332	0.3 h	1.75 m
50	Mill 0.750 ± 0.004 wide $\times 0.375 \pm 0.003$ deep flat.	M	240	F333	0.3 h	1.60 m



THE ROUTE SHEET

8

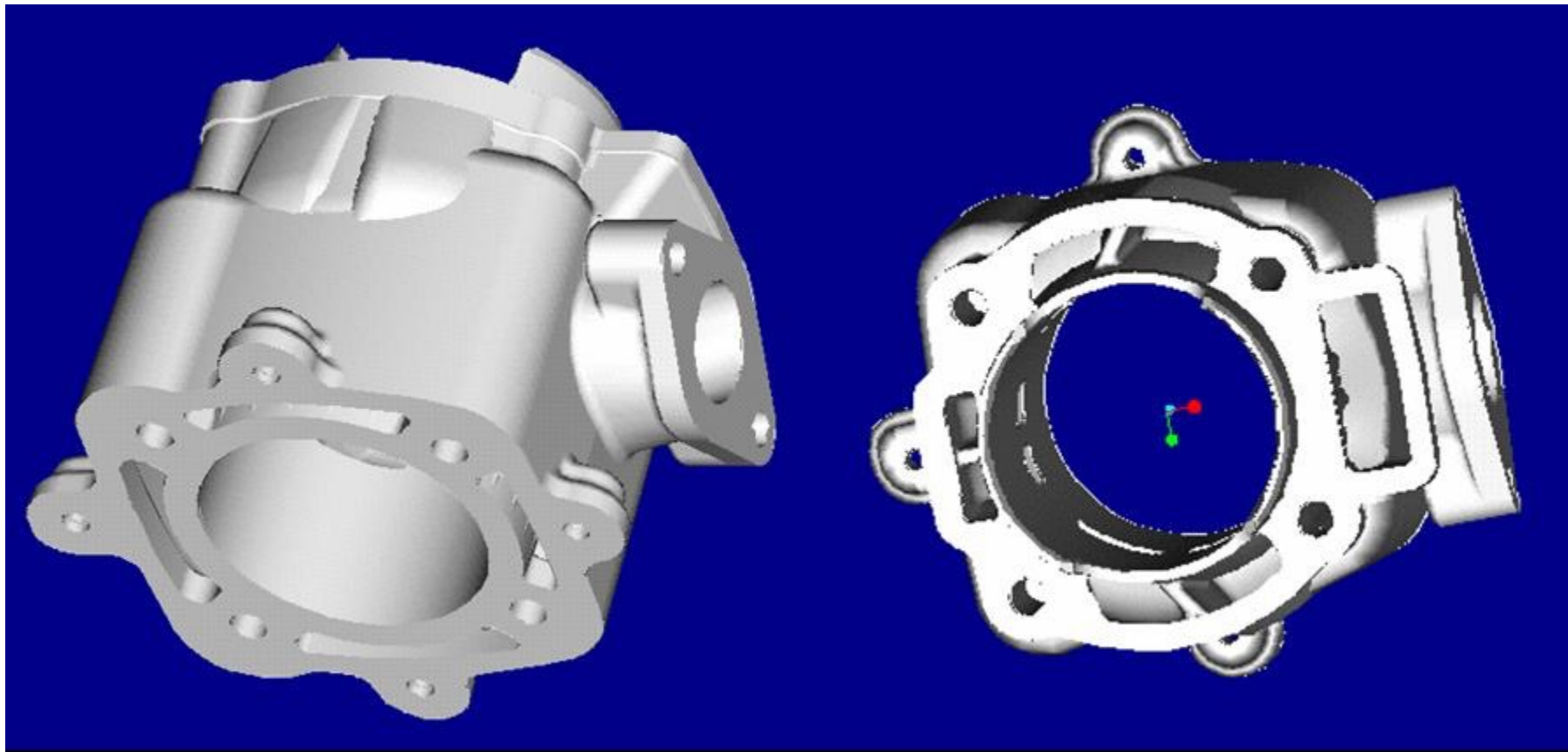
Phase	Position (Fixturing Assembly)	Machined features	Operation sequences	Tool (mill/drill)
1	- Pictures of the assembly 1 	- Pictures of the part  - Pictures of the resulted part (Saved from the simulation) 	1. Facing 2. Contouring N. drilling	tool 1 tool 2 tool N
...	...	...	...	...
N	- Pictures of the assembly N 	- Pictures of the part  - Pictures of the resulted part (Saved from the simulation) 	1. Facing 2. Contouring 3. pocketting N. drilling	tool 1 tool 2 tool 3 tool N

No.	Features	Phase	Tool type	Tool data					a_p	a_e	V_c	n	V_f	K_c	P_c	OK/NO
				D	l1	max a_p	max a_e	...								
1	Facing	1	face mill	5	1	1	1	...	0.5	1	20	100	2	1500	100	OK
...																
No.																



AN EXAMPLE

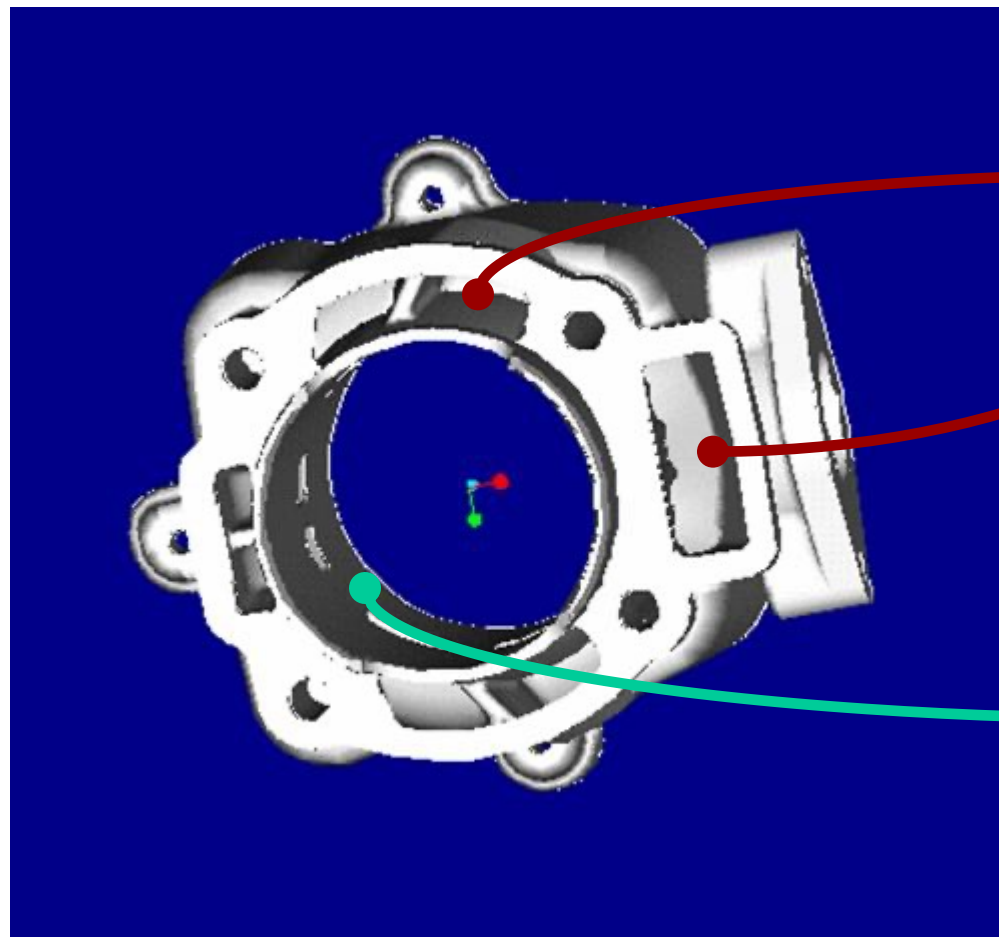
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AN EXAMPLE

10

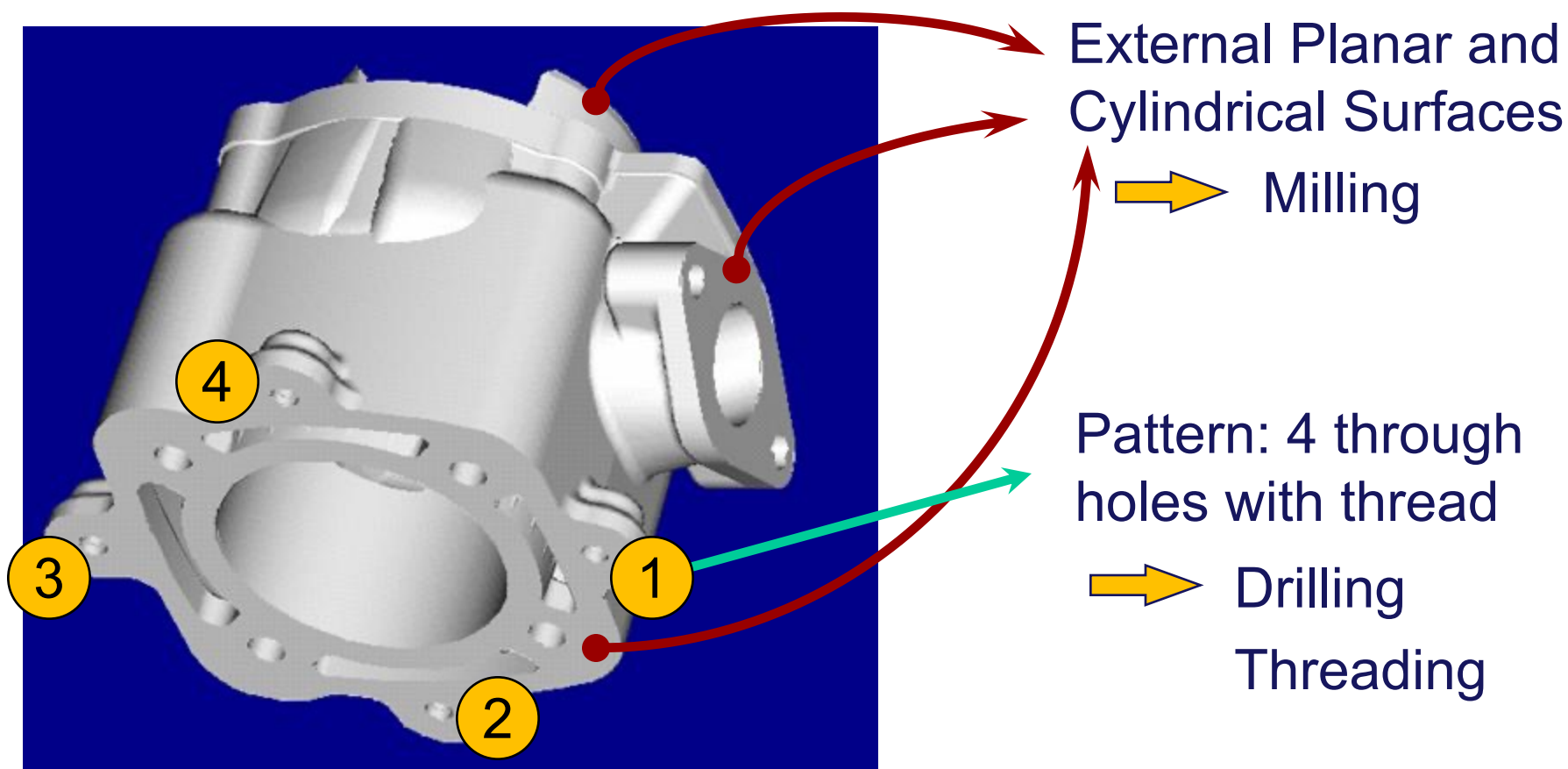


Raw Through Pocket

→ No Operation

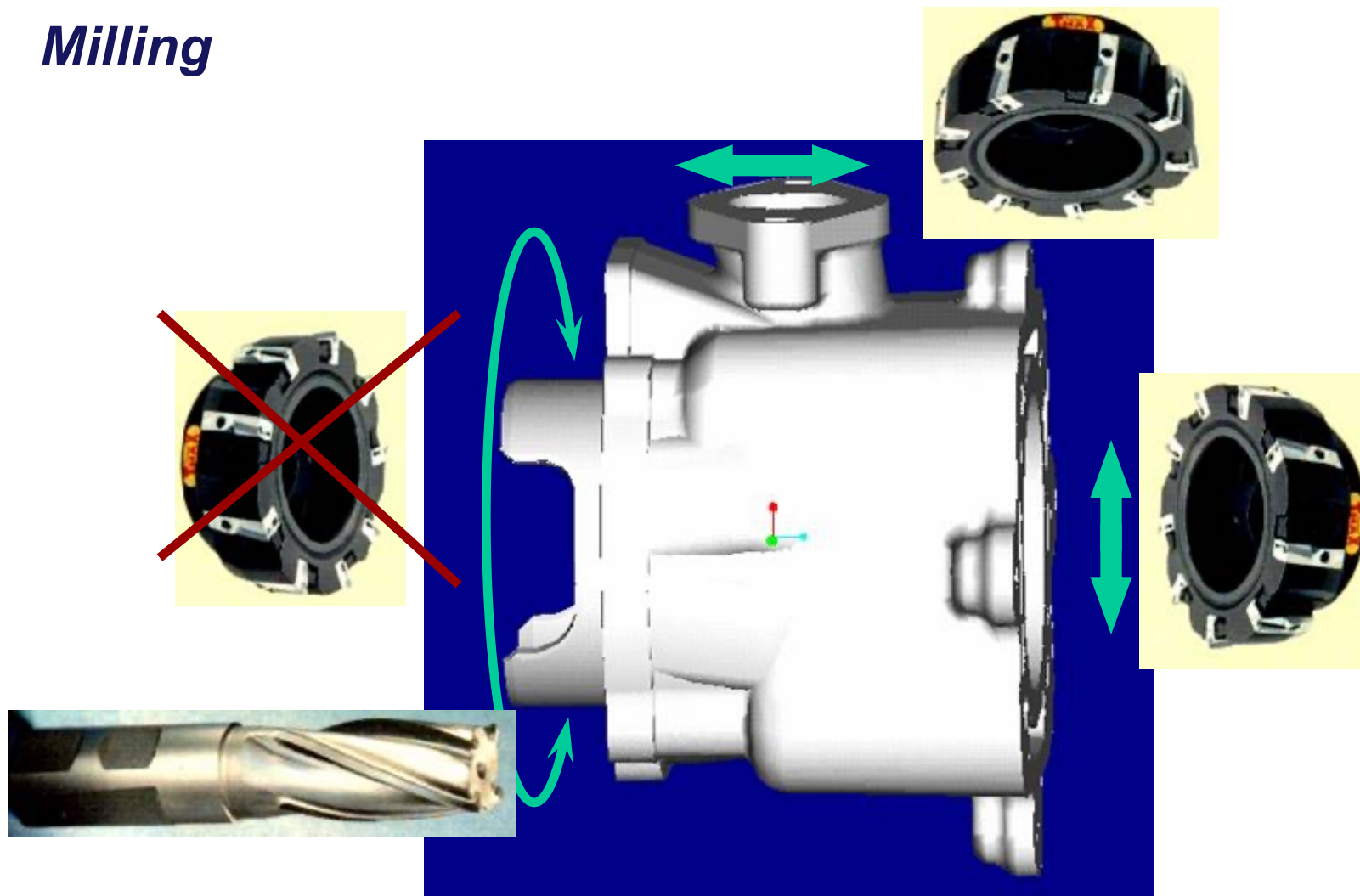
Through Hole

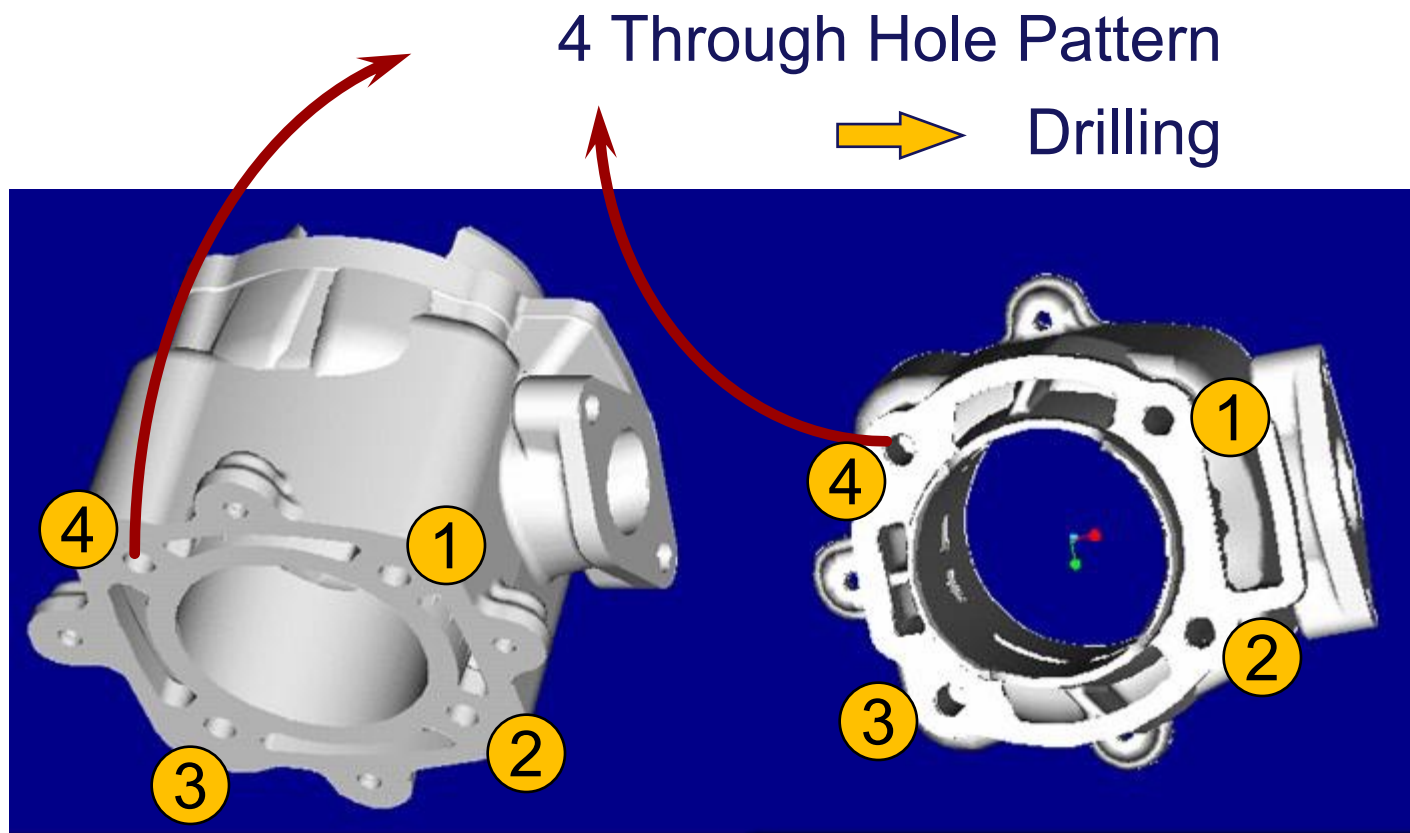
→ Finishing





Milling

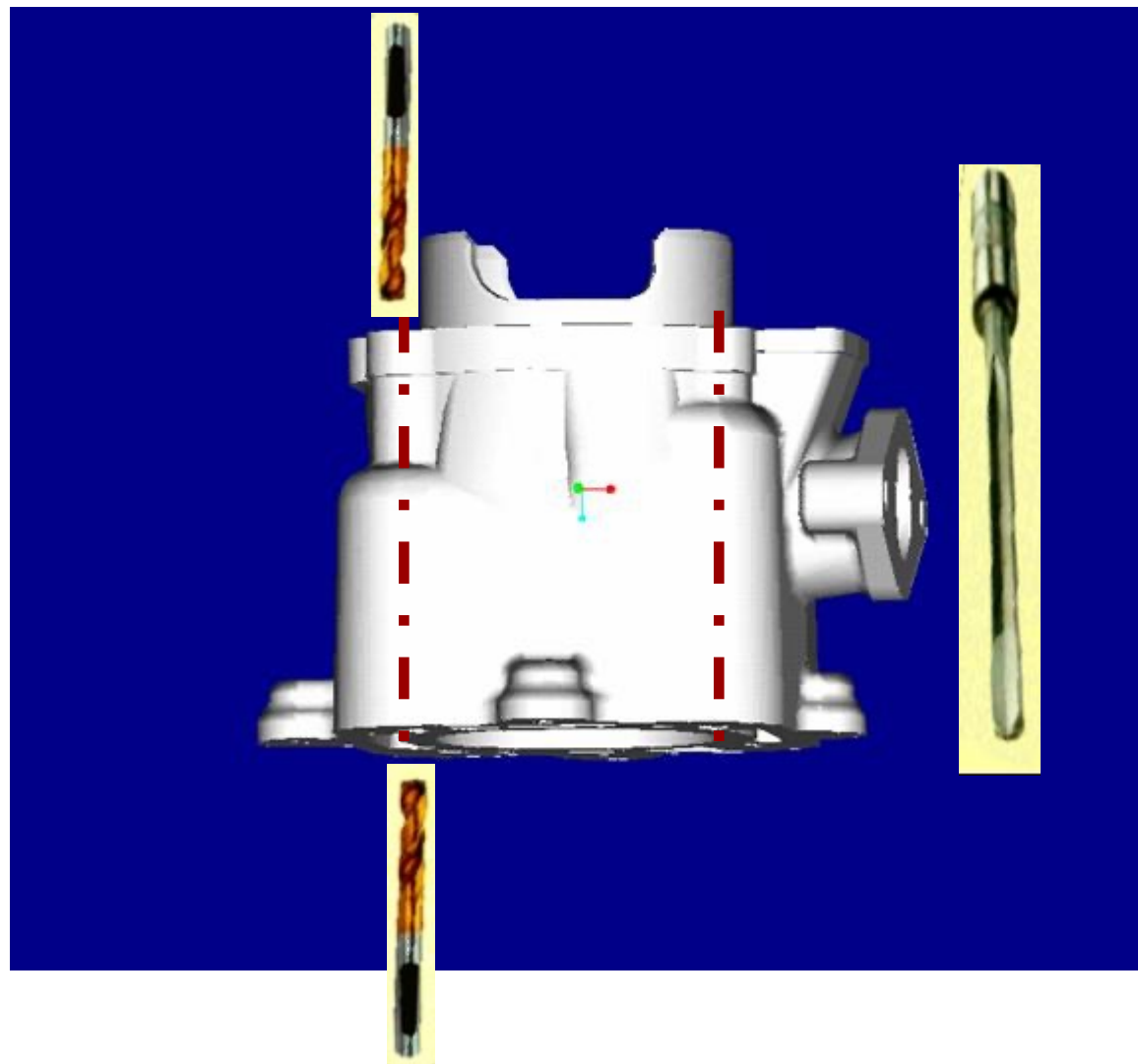




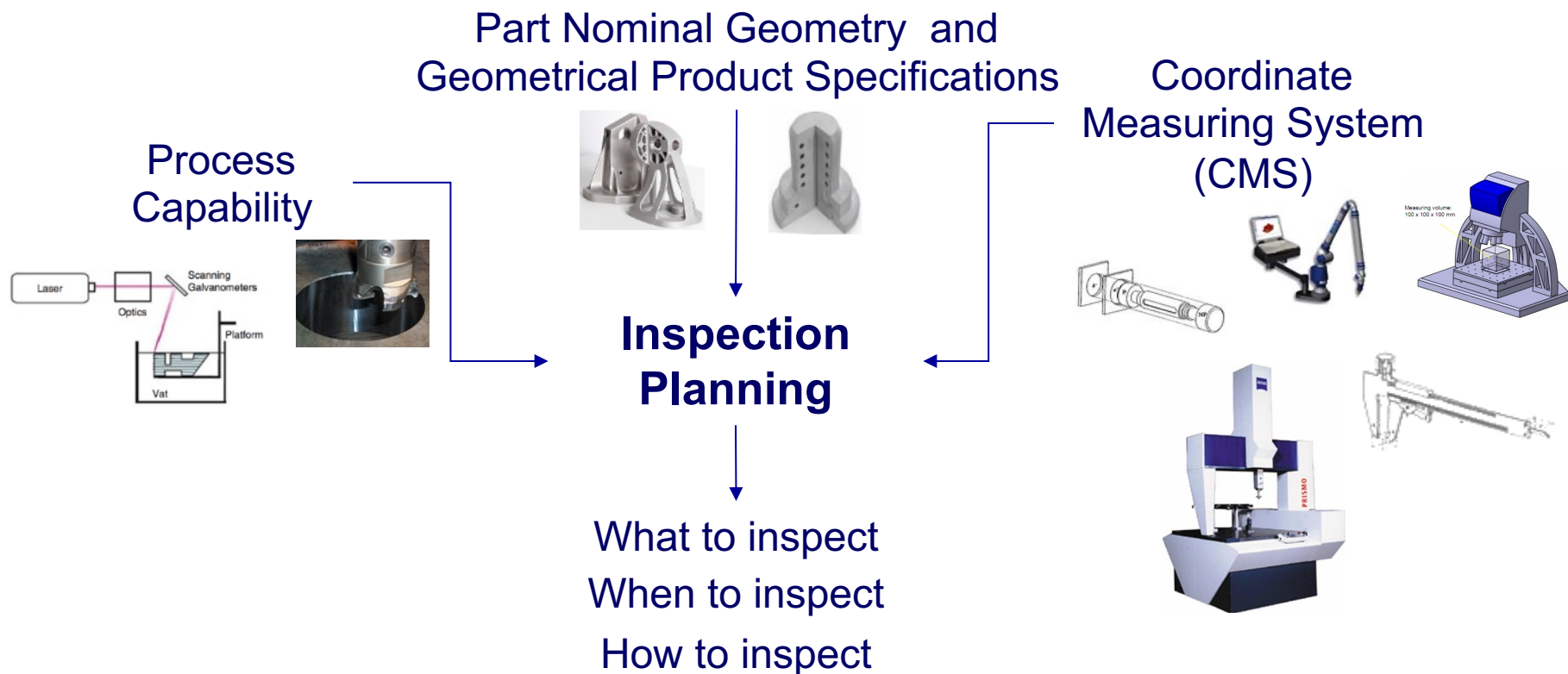


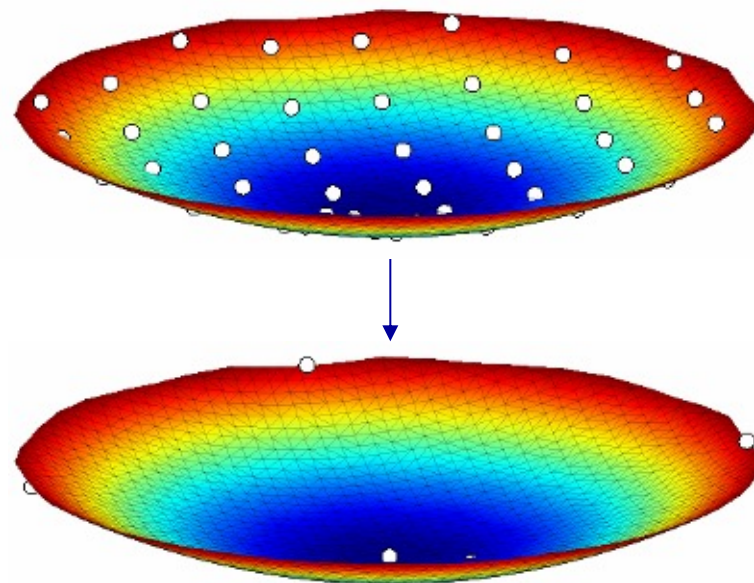
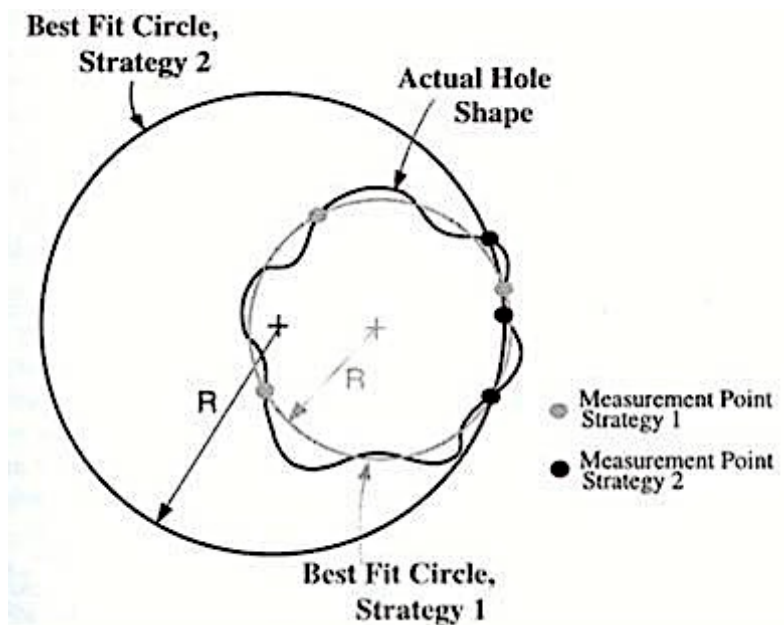
Drilling

Standard
Drilling



Deep
Hole
Drilling

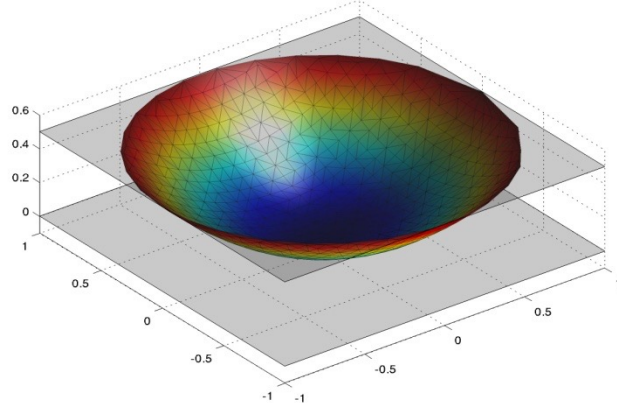






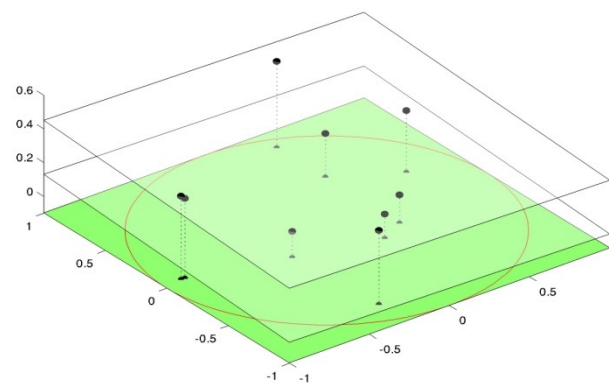
Different sampling strategies may yield different error evaluations.

Real surface



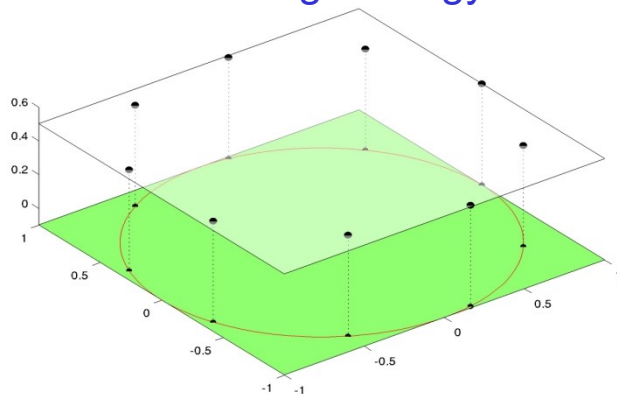
$$e_{real} = 0.5$$

Random strategy



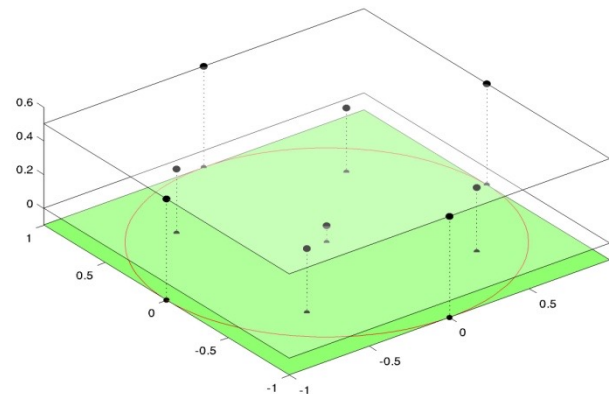
$$\hat{e} = 0.32$$

Wrong strategy



$$\hat{e} = 0$$

Good strategy



$$\hat{e} = 0.5$$



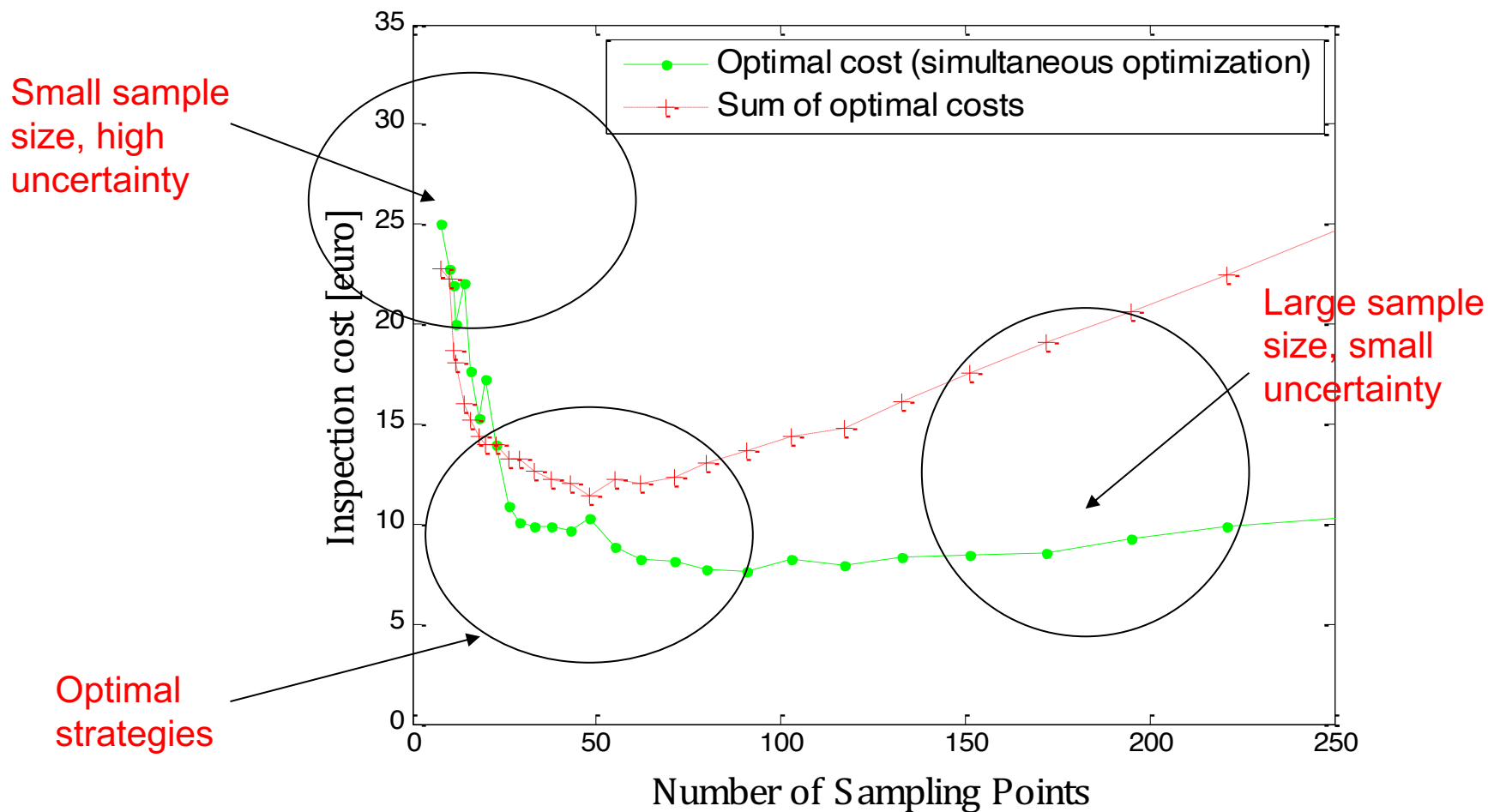
Inspection Cost
 C_I

Inspection Errors
Cost C_E

Cost derived from inspection errors of type A
(indicate as bad a good part)
and type B
(indicate as good a bad part)

Measurement
Cost C_M

Cost arising directly from performing a measurement

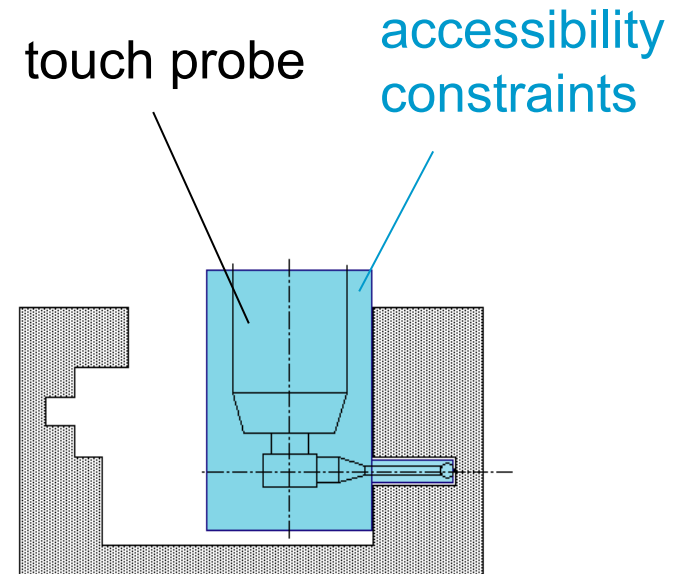
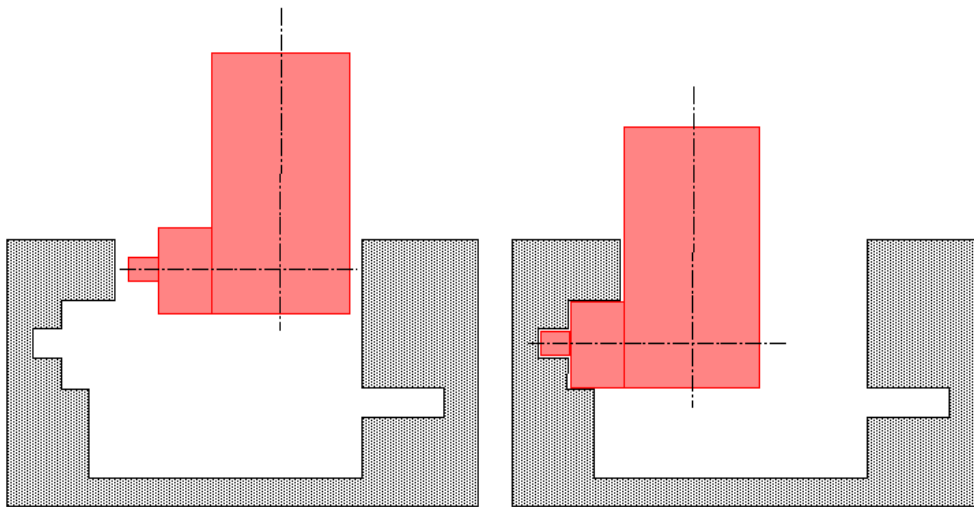




Configure a touch probe means to determine its more suitable **orientation**, **structure**, **dimensions** to inspect the part.

Two main activities:

- Accessibility Analysis
- Clustering and Configuration





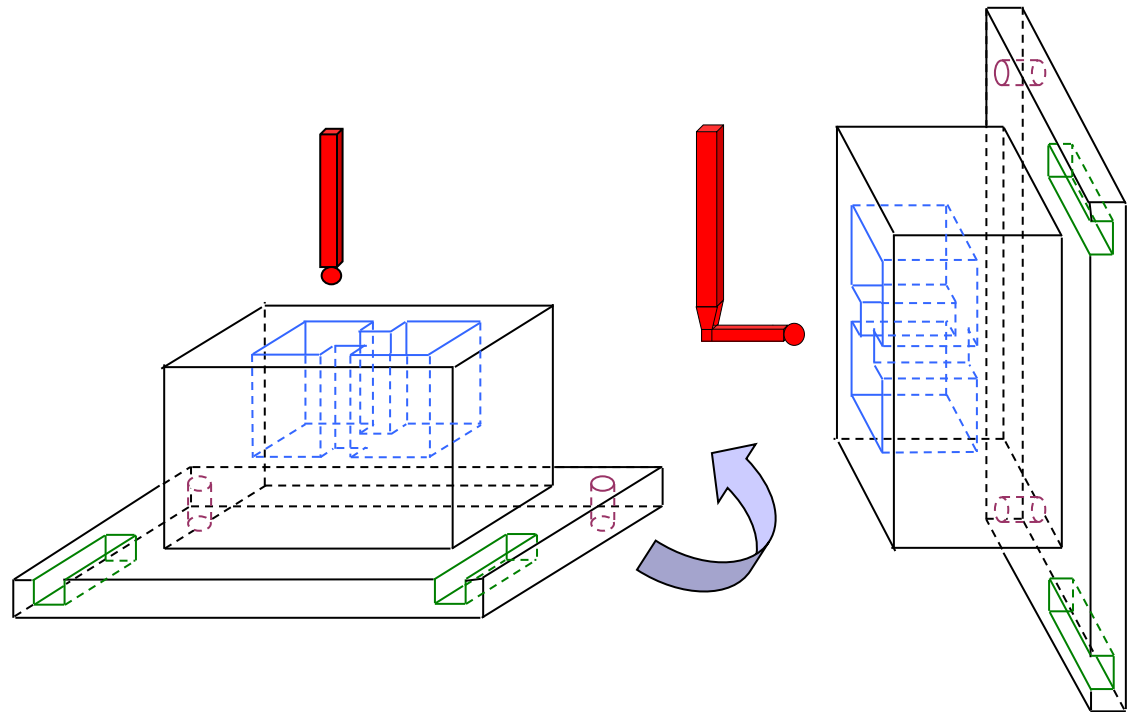
PART ORIENTATION AND FIXTURE CONFIGURATION

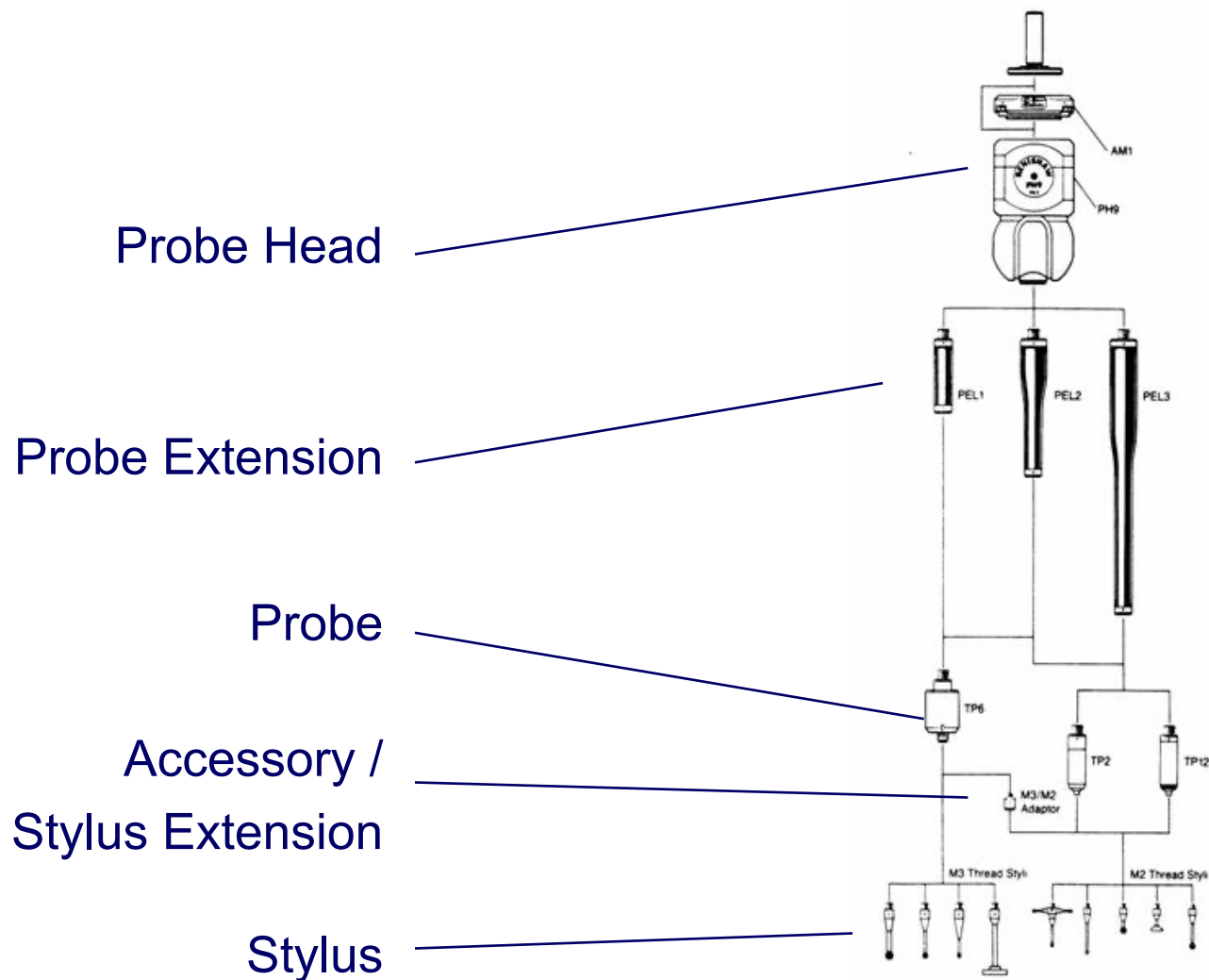
Configure a touch probe
means to determine its more
suitable

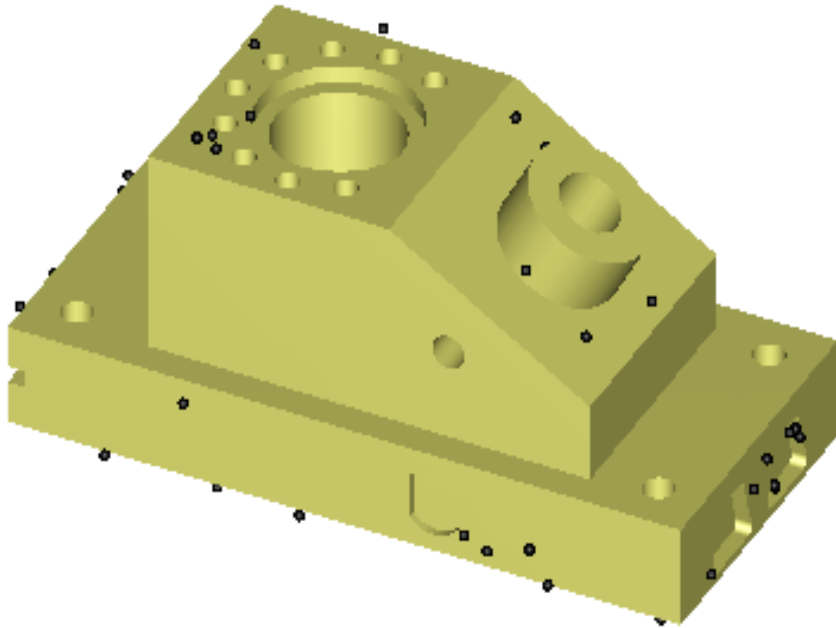
ORIENTATION,
STRUCTURE,
DIMENSIONS
to inspect the part.



PART ORIENTATION







Traveling Salesman Problem

Graph Nodes: measuring points

Arcs: collision free trajectories
between each couple of
points.

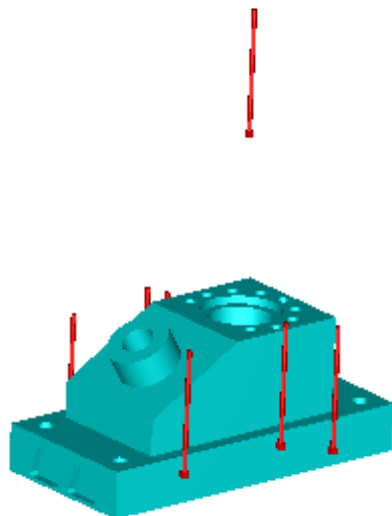
There is not a priori knowledge
of the arc length!

To find an optimal solution to a TSP problem is NP-hard.

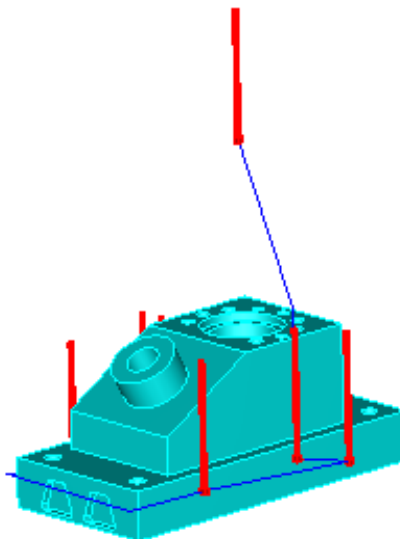
The complexity of an exhaustive search is $N!$

A priori knowledge: $N(N-1)/2$ collision free trajectories.

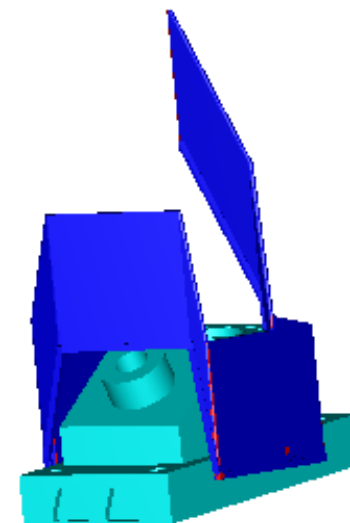
e.g. $N = 50$, 1225 trajectories



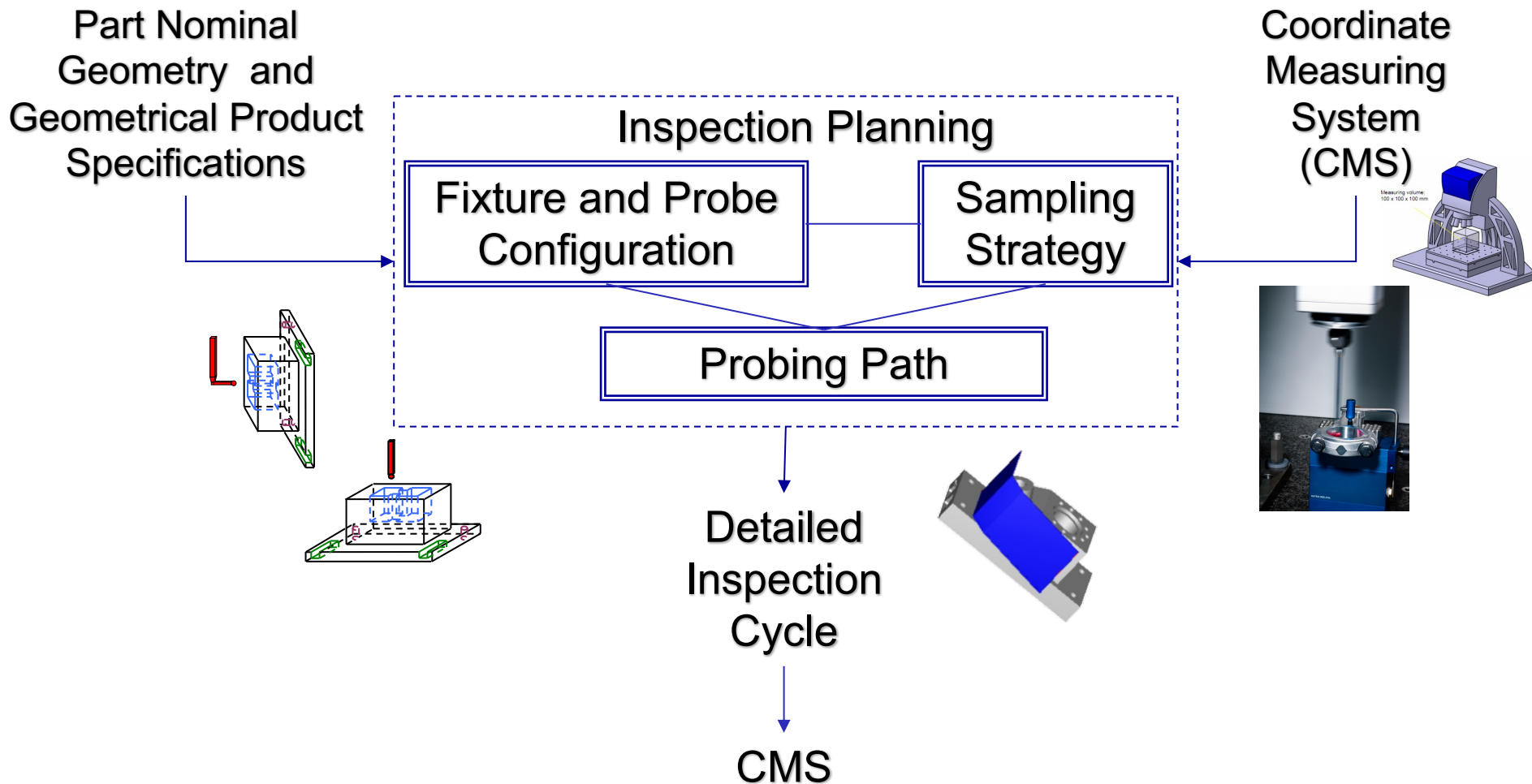
Measuring points on
ANSI/ANC101



Tip path planning



Probe path planning.





KNOWLEDGE-BASED MANUFACTURING

- Shop people knowledgeable in manufacturing processes are gradually retiring.
- Different manufacturing people create different solutions (process plans) to solve the same problem (divergence problem).
- What happens if a new product or a new technology is considered?

An alternative approach to use knowledge is needed:

- During the last several decades, there has been considerable interest in automating the process planning function using computer systems;
- Computer-aided process planning (CAPP) systems provide this alternative.



Computer-aided process planning systems are designed around either of two approaches:

1. Variant systems;
2. Generative systems.

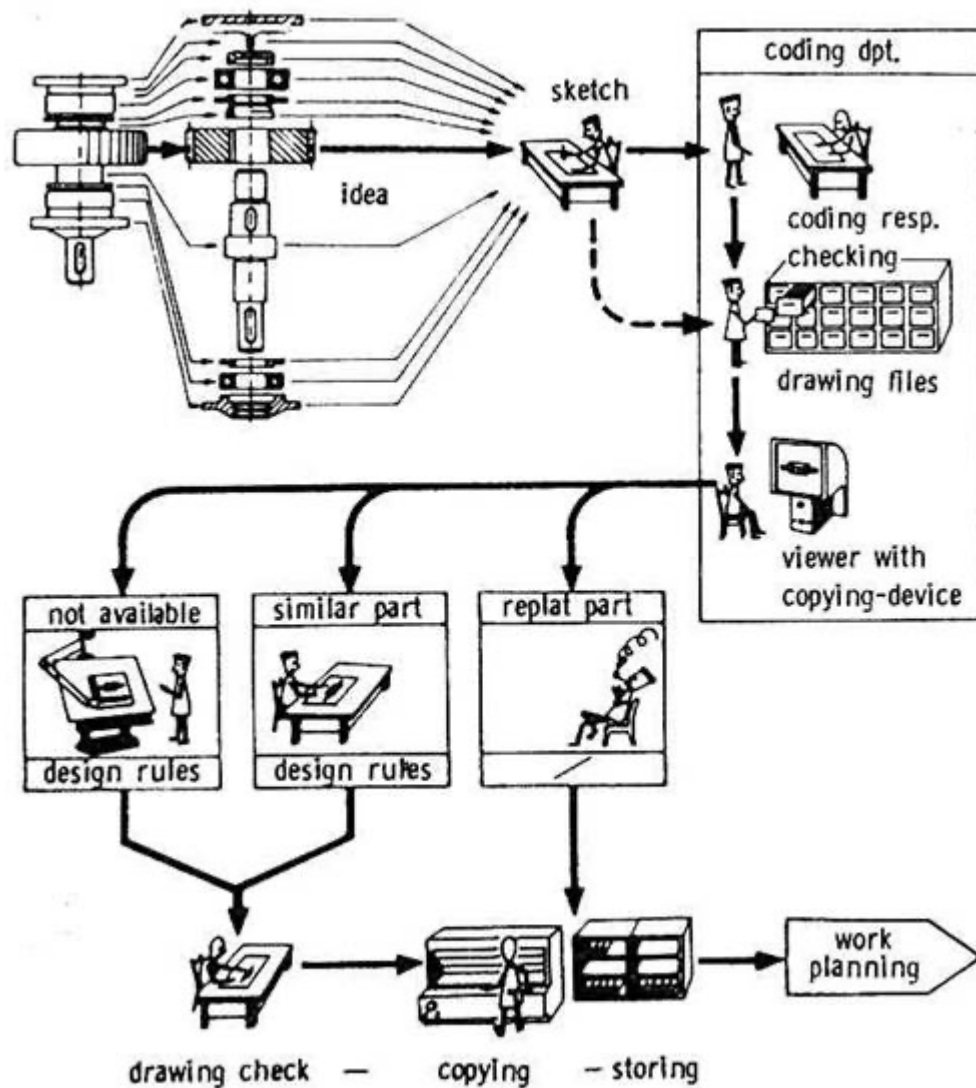
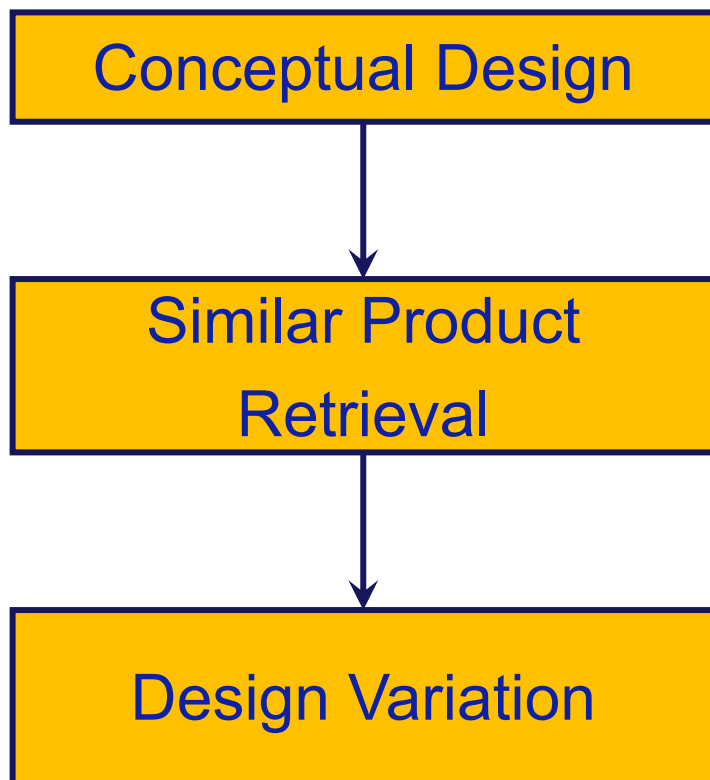


A variant CAPP system is based on an approach called *group technology* (GT):

- Recognizing that although the parts are different, there are families of parts that possess similarities
 - When these similarities are exploited in design and production, overall efficiency is improved.

It is also known as *retrieval CAPP systems*. A standard process plan is stored in computer files for each part code number:

- The standard plans are based on current part routings used in the factory, or on an ideal plan prepared for each family;
- For each new part, the standard plan is edited if modifications are needed.

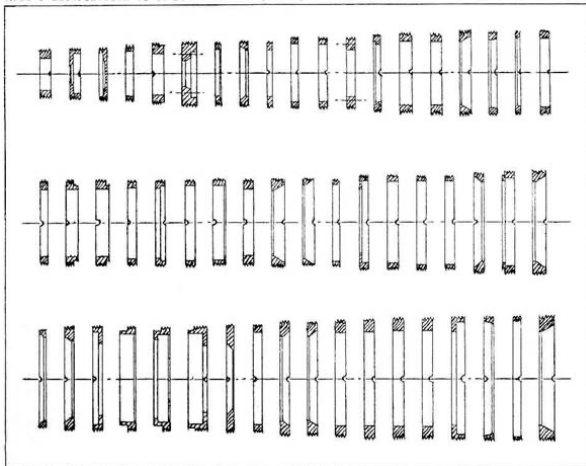




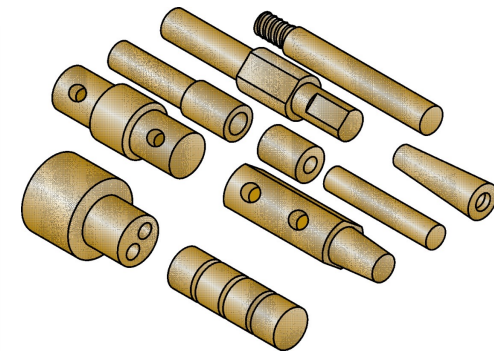
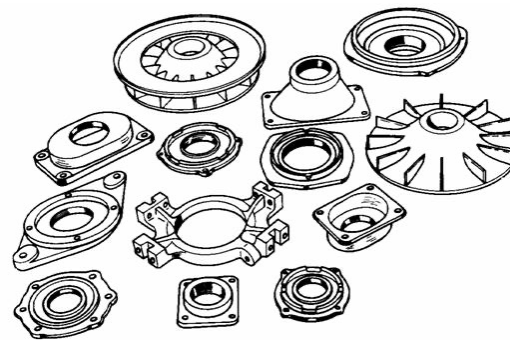
A group of parts that possess similarities in shape and size, or in the processing steps used in their manufacture:

- There are always differences among parts in a family, but the similarities are close enough that the parts can be grouped into the same family.

Design Part Families



Manufacturing Part Families





WAYS TO IDENTIFY PART FAMILIES

1. Visual inspection
 - Using best judgment to group parts into families, based on the parts or photos of the parts.
2. Production flow analysis
 - Using information contained on route sheets to classify parts into manufacturing families.
3. Parts classification and coding
 - Identifying similarities and differences among parts and relating them by means of a coding scheme.



PRODUCTION FLOW ANALYSIS

32

Machine	Pièces N°																																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	
K	✓		✓																	✓			✓		✓											
b		✓					✓			✓		✓	✓					✓						✓				✓				✓				
c	✓		✓		✓										✓		✓												✓		✓					
d		✓					✓					✓	✓											✓			✓								✓	
e								✓						✓		✓																			✓	
f							✓							✓		✓				✓							✓								✓	
g	✓		✓		✓										✓		✓			✓			✓													
h	✓		✓		✓												✓			✓			✓													
i							✓							✓			✓			✓			✓			✓			✓							
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R		✓								✓		✓	✓																						✓	
S				✓		✓			✓		✓																									
T							✓																													

Machine															
	24	25	26	27	31	7	10	18	1	3	5	15			
b	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
d	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
m ⁰	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
n	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
R	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
a												✓	✓		

Machine	Pièces N°																																		
	2	12	13	24	27	31	7	10	18	1	3	5	15	17	20	23	25	29	4	6	9	11	21	28	35	30	32	33	8	14	19	22	26	16	34
b	✓	✓	✓	✓	✓	✓	✓	✓	✓	} Gr1																									
d	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
m ⁰	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
n	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
R	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
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PARTS CLASSIFICATION AND CODING

Most classification and coding systems are one of the following:

- Systems based on part design attributes;
- Systems based on part manufacturing attributes;
- Systems based on both design and manufacturing attributes.

Design attributes

Major dimensions
Basic external shape
Basic internal shape
Length/diameter ratio
Material type
Part function
Tolerances
Surface finish

Manufacturing attributes

Major process
Operation sequence
Batch size
Annual production
Machine tools
Cutting tools
Material type



PARTS CLASSIFICATION AND CODING

Classification and coding system according to Opitz.

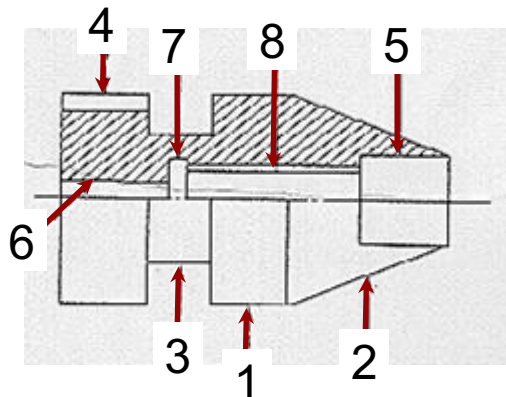
		Form Code					Supplementary Code							
		1st digit Part class	2nd digit Main shape	3rd digit Rotational surface machining	4th digit Plane surface machining	5th digit Auxiliary holes, gear teeth, and forming	Digit							
							1	2	3	4				
0	Rotational parts	$\frac{L}{D} < 0.5$	External shape, external shape elements	Internal shape, internal shape elements	Plane surface machining	Auxiliary holes	Dimension	Material	Original shape of raw material	Accuracy				
1		$0.5 < \frac{L}{D} < 3$												
2		$\frac{L}{D} > 3$				Gear teeth								
3		$\frac{L}{D} < 2$ with deviation	Main shape	Rotational machining, internal and external shape elements		Auxiliary holes, gear teeth, and forming								
4		$\frac{L}{D} \leq 2$ with deviation												
5		Special												
6	Nonrotational parts	$\frac{A}{B} < 3, \frac{A}{C} > 4$ Flat parts	Main shape	Principal bores	Plane surface machining	Auxiliary holes, gear teeth, and forming								
7		$\frac{A}{B} > 3$ Long parts	Main shape											
8		$\frac{A}{B} < 3, \frac{A}{C} > 4$ Cubic parts	Main shape											
9		Special												

COMPOSITE PART CONCEPT

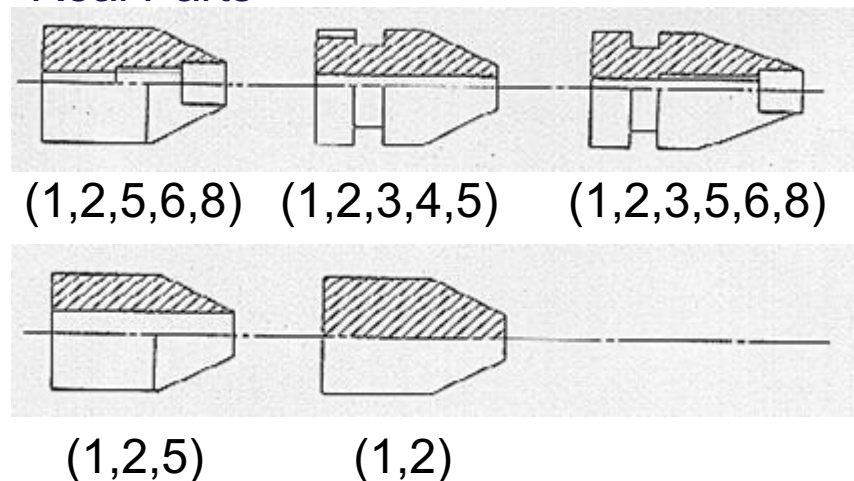
A composite part for a given family is a hypothetical part that includes all of the design and manufacturing attributes of the family:

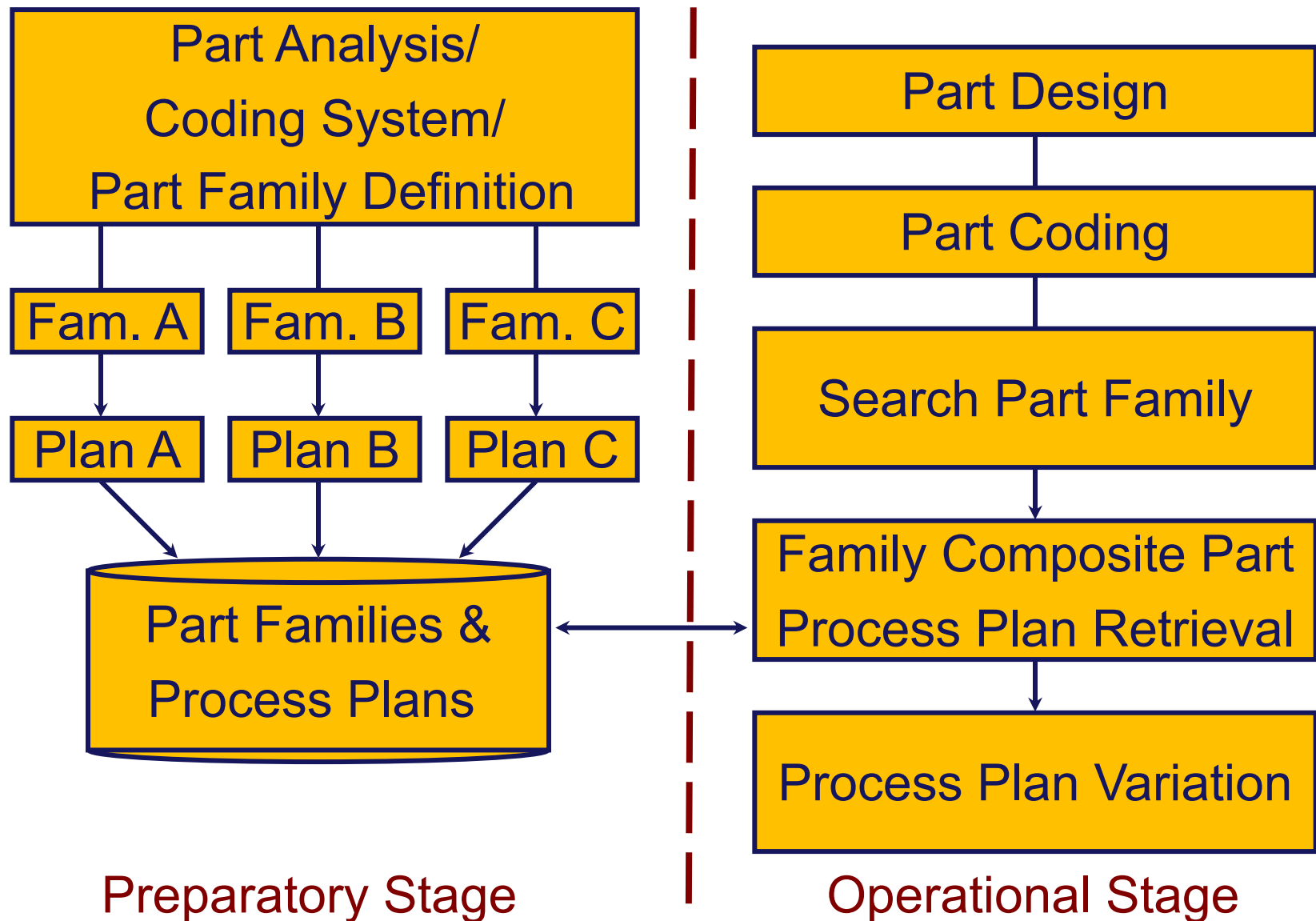
- An individual part in the family will have some of the features of the family, but not all;
- A process plan for the part family would consist of operations required to make the composite part:
 - Such a plan would be able to produce any family member, by omitting operations corresponding to features not possessed by that part.

Composite Part



Real Parts







GENERATIVE CAPP SYSTEMS

- Rather than retrieving and editing an existing plan, the process plan is created using systematic procedures that might be applied by a human planner.
- In a fully generative CAPP system, **the process sequence is planned without human assistance and without predefined standard plans.**
- Designing a generative CAPP system is a problem in ***expert systems*** - computer programs capable of solving complex problems that normally require a human with years of education and experience.



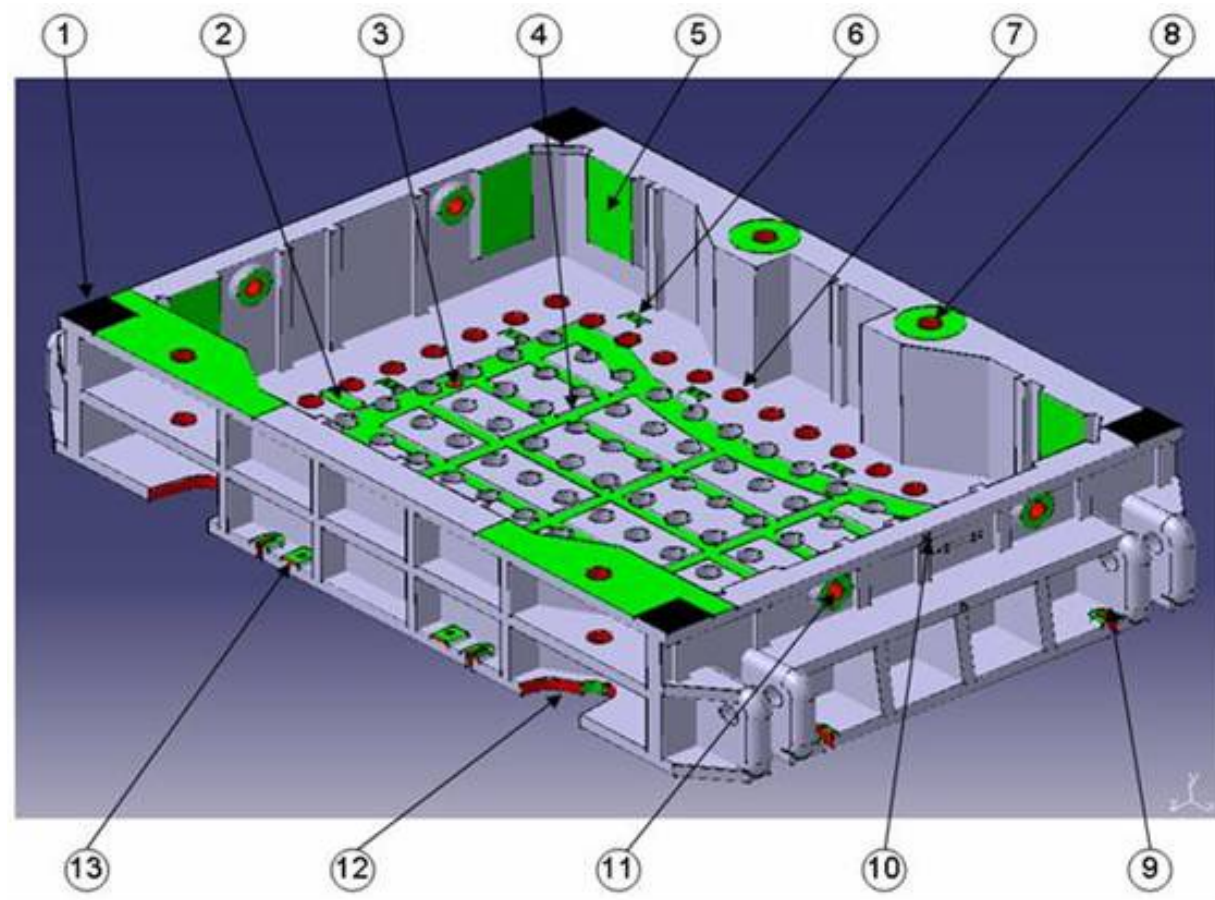
COMPONENTS OF AN EXPERT SYSTEM

- **Knowledge base** - The technical knowledge of manufacturing and logic used by process planners must be captured and coded in a computer program.
- **Computer-compatible part description** - The description must contain all the pertinent data needed to plan the process sequence.
- **Inference engine** - The algorithm that applies the planning logic and process knowledge contained in the knowledge base to a given part description.



COMPUTER-COMPATIBLE PART DESCRIPTION

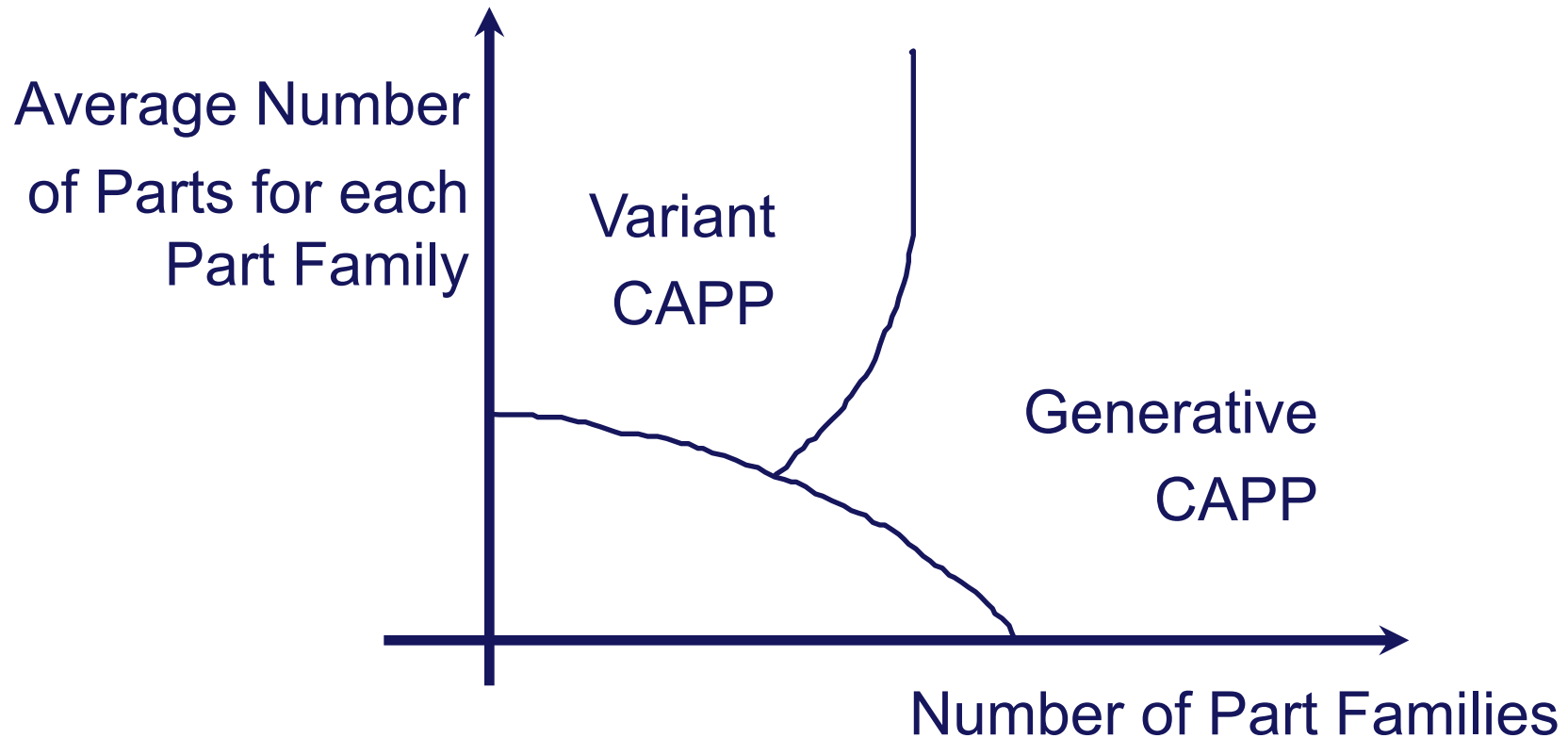
- Form Feature and Form Feature Recognition

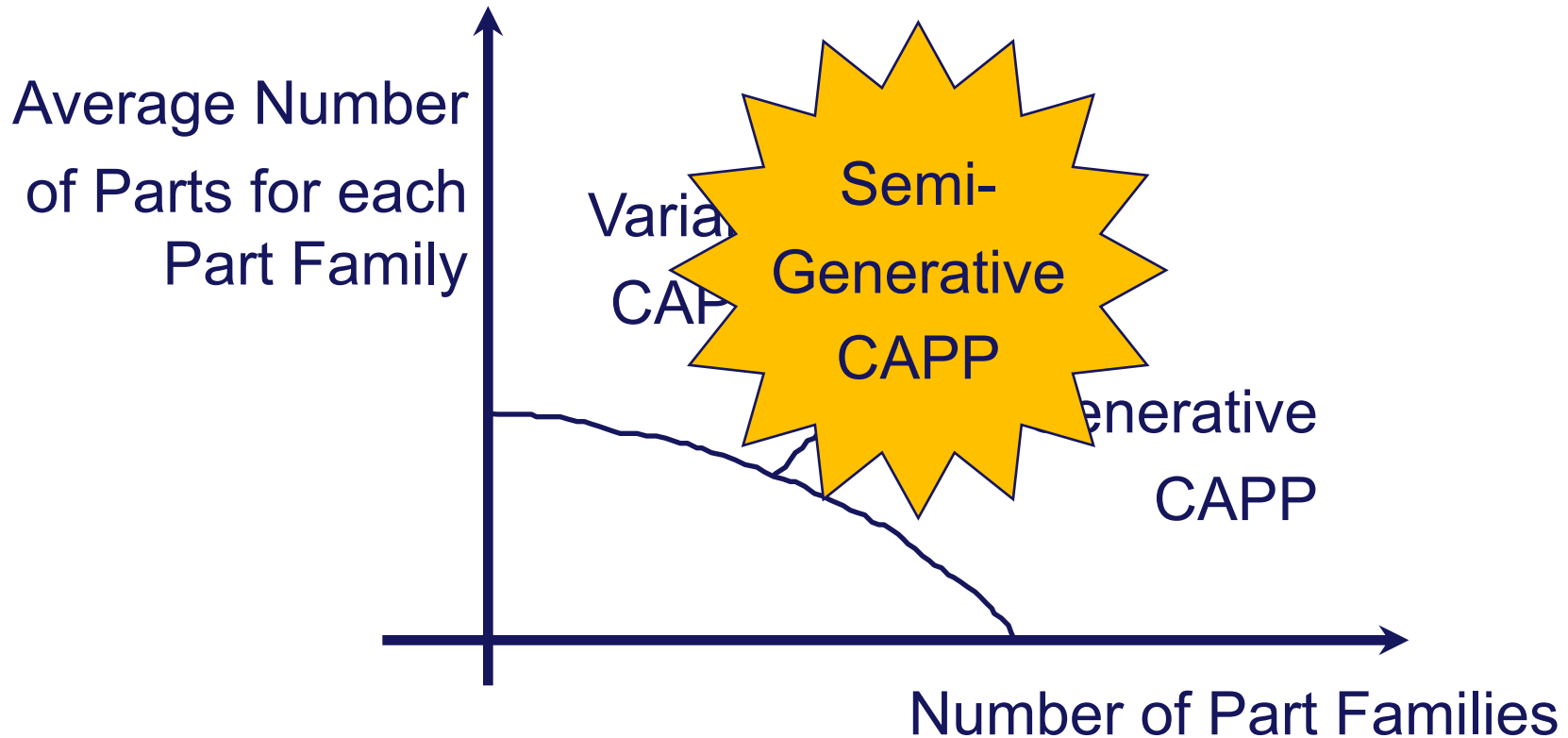




Critical Factors

- ▶ Manufacturing System Components
- ▶ Production Volume
- ▶ Number of Part Families







Semi-Generative CAPP

- ▶ Variant approach for process plan retrieval + Generative approach for standard process plan modification
- ▶ Generative approach for macro planning + Variant approach for detailed plan definition.



- **Process standardization and optimization** – CAPP leads to more logical and consistent process plans than traditional process planning.
- **Better re-use of knowledge and improved training.**
- **Increased productivity of process planners**, by reduced lead time to prepare process plans.
- **Decreased time and cost for process plan maintenance.**
- CAPP programs can be interfaced with other application programs, such as cost estimating, work standards, and others.